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Tailoring microstructure of additive friction stir-deposited Al–Mg alloy through post-processing deformation treatment for enhancing mechanical performance

Wenshen Tang, Xinqi Yang *, Ruilin Wang, Ting Luo

Tianjin Key Laboratory of Advanced Joining Technology, School of Materials Science and Engineering, Tianjin University, Tianjin, China

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ABSTRACT

This study explored the influence of the amount of post-processing deformation on the microstructure and mechanical properties of Al-5083 fabricated by the additive friction stir deposition (AFSD) process. The findings indicated that, through post-processing deformation treatment, the tensile properties achieved were comparable to those of wrought 6061-T6 Al alloy.

1. Introduction

Many aluminum alloys of engineering significance, particularly Al–Mg alloys widely used in marine and offshore applications, are susceptible to elemental burnout and have high thermal fracture sensitivity [1,2]. These characteristics make them prone to porosity and cracking during solidification, posing challenges for their use in fusion-based additive manufacturing processes [3,4]. To enhance the printing formability of Al–Mg alloys, researchers have explored alloying methods, such as the addition of Zr [3] and Sc [5] elements. While these methods significantly improved the tensile properties of Al–Mg alloys, they have not been entirely successful in eliminating defects like porosity and cracking [3,5]. Moreover, a significant grain-preferred orientation was observed along the direction of deposit thickness accumulation [6]. This, combined with the aforementioned defects, resulted in considerable anisotropy in the fatigue properties of the alloyed components [6]. An alternative approach is the use of solid-state friction-based additive manufacturing. This method can effectively circumvent the defects associated with fusion-based additive manufacturing of commercial Al alloys, as it does not involve the melting and solidification process of the material [7,8].

Additive friction stir deposition (AFSD) stands as one of the most advanced solid-state friction-based additive manufacturing processes. Its unique ability to freely control deposition paths, coupled with its potential to address the challenging printability of commercial Al alloys, has attracted significant attention. As a result, the development and application of AFSD have been progressing rapidly in recent years [8,9]. However, the yield strength in the longitudinal direction of the Al–Mg alloy parts produced by AFSD ranged from 55.3% to 77.7% of the starting materials (AA5083-H131 and 5083-H112 consumable bars) [10,11]. The observed softening in the as-built Al–Mg alloy was primarily attributed to the higher degree of grain recrystallization induced by the AFSD process, coupled with a significantly lower dislocation density in the as-deposited sample compared with the base material [10, 11]. Consequently, the contributions of both Orowan and dislocation strengthening are diminished considerably. To enhance the performance and thereby broaden the range of applications of Al–Mg alloy components fabricated via AFSD, the exploration of appropriate post-processing methods emerges as a crucial factor.

As Al–Mg alloys are not age-hardenable, solid solution strengthening and strain hardening are their primary strengthening methods [3,4]. Solid solution strengthening requires adding additional alloying elements [3]. Strain strengthening, such as pre-stretching [12] and rolling [13], are all post-processing deformation treatment of strengthening methods that have received much attention. These readily applicable strengthening methods can lead to significant performance improvements in the industry without requiring changes to the alloy composition [12,13]. Therefore, this study introduced a compression deformation process design to investigate the impact of varying deformation degrees on the microstructure and mechanical performance of thick-walled Al–Mg alloy structures fabricated via AFSD. The experimental findings are anticipated to offer fresh insights and a theoretical

* Corresponding author. Materials Processing Engineering Department, Tianjin University, Tianjin, China.
E-mail address: xqyang@tju.edu.cn (X. Yang).

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foundation for regulating the microstructure and properties of large-sized, complex Al–Mg alloy components.

2. Material and methods

In this study, the 5083-Al alloy extruded bar was the consumable material for the deposition experiment, while the 5 mm thick 5083-H112 wrought plate was the substrate. The nominal chemical composition of the starting materials is Al-4.59Mg-0.66Mn-0.29Si-0.14Zn-0.11Fe-0.11Cr-0.04Ti-0.03Cu (wt.%). The deposition experiment was conducted using the first generation of solid-state friction-based deposition equipment that Tianjin University developed. The printing tool was made of H13 tool steel with a diameter of 38 mm, and its surface was machined with two 2.5-mm-tall protrusions in a symmetrical arrangement. A six-layer, single-pass deposit was produced using optimized parameters with a tool rotation speed of 350 r/min, a tool traverse speed of 200 mm/min, a feeding pressure of 10 kN, and a pre-set layer thickness of 1.5 mm. The length of the final deposit was 250 mm center-to-center.

The squares of 16 mm × 50 mm × 14 mm were taken from the final deposit for compressive deformation tests. The deformation treatment tests were performed with 3%, 6% and 9% compressive deformation amount (DA=$((h_0-h_1)/h_0)$, Fig. 1(b)) using a tensile testing machine (range 0–500 kN) in the direction of the accumulated thickness of the deposited layer. Before commencing the compression tests, a calibration experiment was conducted using specimens with uniform dimensions. This initial procedure was critical to determining the relationship between the displacement of the tensile machine crosshead and the actual compression of the sample. With the crosshead set to progress at a rate of 2.5 mm/min, specimens were fabricated to achieve specific compressive deformation levels based on the previously established correlation. The microstructural features at the first and fifth interfaces (referred to as the top and bottom regions in this study) of the as-deposited and post-processing deformation-treated samples were examined by electron backscattering diffraction (EBSD, Zeiss Sigma 500-SEM with Symmetry CMOS-based detector (Oxford Instruments)). A scanning step of 0.55 μm was used with a 200 × 100 mm scanning area. Grain size, recrystallization ratio and geometrically necessary dislocations (GNDs) density at the examined regions were calculated and counted by comparison based on the obtained EBSD results. The methodology and procedure for calculating the GNDs were described in detail in a previous publication [14]. In this study, slip systems along the <110> direction on {111} planes were considered.

Hardness and tensile tests were performed on the specimens in the as-deposited and deformation-treated states to evaluate further the homogeneity and mechanical properties of the deformed microstructure. Hardness was measured using a digital display durometer (HVS-1000) and a load of 0.98 N for 15 s. The tensile specimens (Fig. 1(c)) were cut along the width direction of the deposited samples, and the tensile tests were performed using an electronic universal tensile testing machine (CSS-44100) at a constant crosshead speed of 2.5 mm/min.

3. Results and discussion

During AFSD, the consumable feedstock bar experienced frictional heat and shearing forces imparted by the printing tool. As a result, a dynamic recrystallization ratio of 83.9%–85.7% was achieved. Significant grain refinement was observed in the as-deposited state, with the refined grain size being 6.1–8.0% of the starting material (Fig. 2 and Table 1). In final deposits of heat-treatable aluminum alloys produced by friction stir-based deposition processes, pronounced grain refinement was also observed [15].

Upon introducing a minor extent of plastic deformation, the grain morphology exhibited minimal alteration, with no pronounced flattening observed. When subjected to 3% plastic deformation, the grain size remained largely consistent with that of the as-deposited condition. The percentage of LABs increased to 22.1–29.1%, and the substructured grain rate increased to 20.4–34.0%. As the amount of deformation increased to 6%, the grain size remained close to the as-deposited state. However, the LABs fraction surged to 30.9–41.7%, and the percentage of substructured grains climbed to 38.5–60.0%. With the application of 9% plastic deformation, discernible grain deformation emerged relative to the as-deposited state. Localized regions exhibited grain fragmentation, leading to a marginal reduction in grain size. The LABs fraction expanded to 37.5–42.9%, and the rate of substructured grains grew to 43.7–59.8%.

The proportion of LABs, substructured and deformed grains increased dramatically with the escalation in the degree of plastic deformation. Concurrently, the density of dislocations within these grains also intensified (Fig. 3). The distribution of the GNDs remained largely uniform at both the top and bottom of the deposited layer. Upon increasing the deformation amount to 3%, the mean GNDs density of the

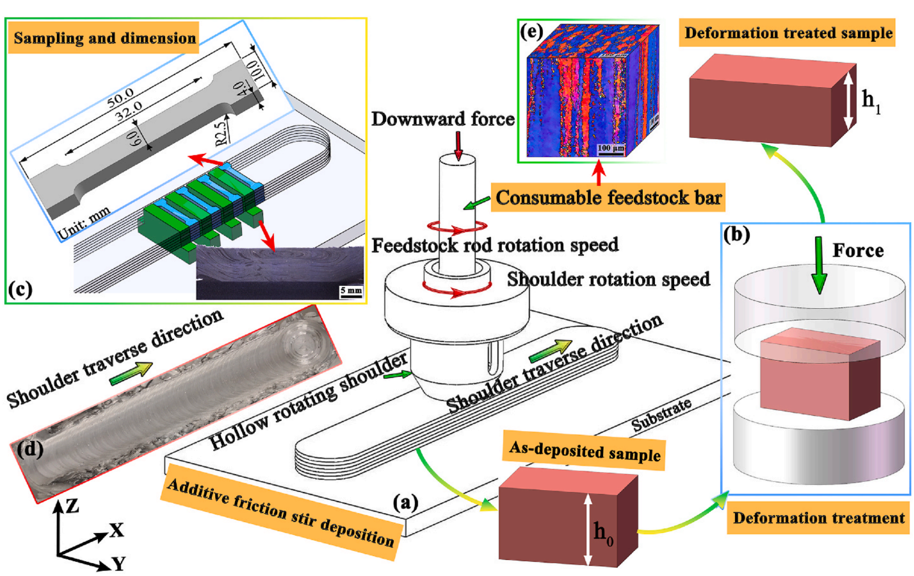

Fig. 1. Schematic illustration of (a) AFSD and (b) post-deformation treatment processes; (c) positions and dimensions of tensile specimens extracted in the Y-direction of (d) final deposits; (e) three-dimensional map of the inverse pole figures (IPF) distribution of the starting material. The inset in the bottom right corner of Fig. 1(c) displays the macro view cross-section of the as-deposited sample.

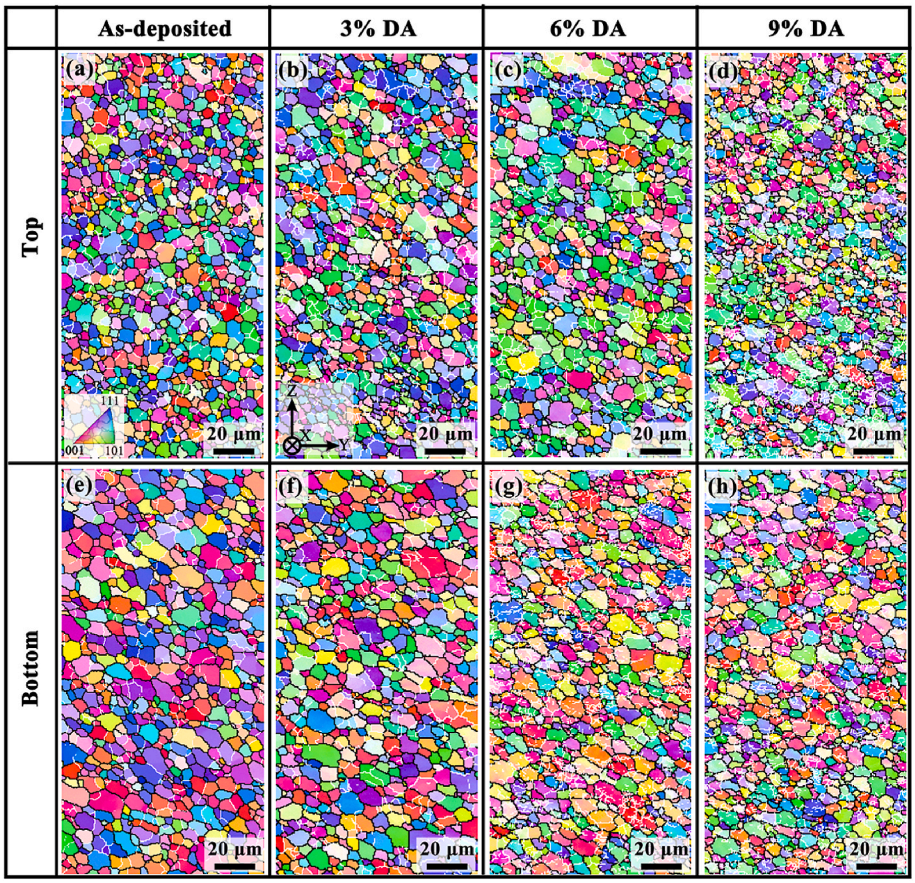

Fig. 2. IPF distribution maps with superimposed grain boundaries (HABs are depicted by black lines, while LABs are depicted by white lines) for top and bottom regions of as-deposited and post-processing deformation-treated samples.

Table 1
Grain size, grain boundary ratio and percentage of different grain types at the examined regions of the as-deposited and deformation-treated samples.

Examined regions		Grain size/μm	LABs/%	Percentage of different grains/%		
				Recrystallized	Substructured	Deformed
Starting material		95.9 ± 63.7	66.6	10.2	69.8	20.0
As-deposited	Top	5.9 ± 2.1	20.4	83.9	13.8	2.3
	Bottom	7.7 ± 3.0	19.4	85.7	13.5	0.8
3% DA	Top	6.8 ± 2.6	29.1	59.5	34.0	6.5
	Bottom	7.9 ± 3.2	22.1	77.1	20.4	2.5
6% DA	Top	7.1 ± 2.9	30.9	55.0	38.5	6.5
	Bottom	7.6 ± 2.5	41.7	22.7	60.0	17.3
9% DA	Top	5.7 ± 2.2	42.9	19.8	43.7	36.5
	Bottom	6.8 ± 2.3	37.5	25.2	59.8	15.0

samples rose to twice that of the as-deposited sample. As the deformation was further augmented to 6% and 9%, the average GNDs density surged to 4.2 and 5.6 times the as-deposited state, respectively.

The as-deposited sample exhibited a relatively uniform hardness, reaching an average of 97% of the hardness of the consumable base material (Fig. 4(a) and (b)). After applying a certain degree of compressive strain, the hardness distribution within the deposited layer maintained uniformity. However, the hardness of the substrate was slightly lower than that of the deposits, a discrepancy likely attributable to the differing strain-hardening capabilities between the deposited layer and the substrate. Compared with the as-deposited specimen, the average hardness values of the post-processing deformation-treated samples increased by 15.2%, 26.2% and 32.8%, respectively, with the increase of the deformation amount.

The tensile test results revealed that the tensile strength, yield strength, and elongation at the break of the deposited specimen could reach 93.7%, 74.3%, and 212.9% of the starting material, respectively (Fig. 4 and Table 2). For ease of comparison, only the deposited layer was subjected to tensile tests, excluding the substrate. Following the application of compressive deformation, there was a significant enhancement in tensile properties, with the yield strength escalating by more than 73.9%, while the elongation at break consistently exceeded 18.4%.

Remarkably, these tensile properties surpassed those of the wrought alloys 5083-H112 and 6061-T6 (Table 2). The tensile strength and yield strength obtained at 6% and 9% deformation were superior to those at 3% deformation. Although the highest average strength was achieved at a deformation amount of 9%, the standard deviation of the test values was also the highest. This high standard deviation indicates a greater spread in the data, suggesting a relative lack of uniformity in the

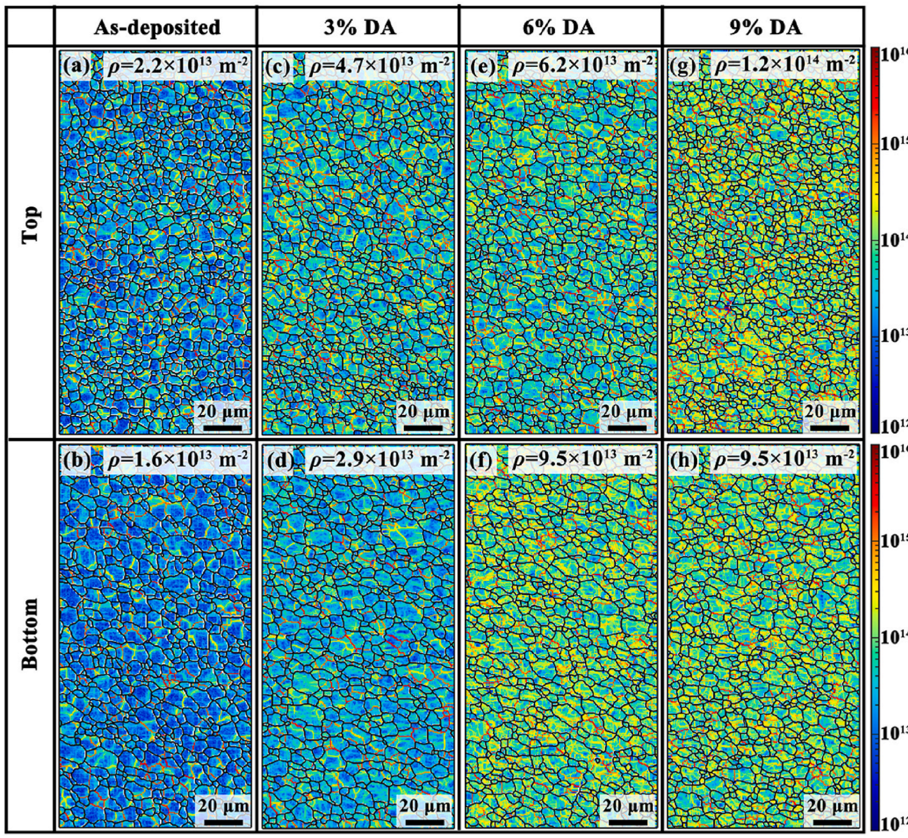

Fig. 3. GNDs density distribution maps for the top and bottom regions of as-deposited and post-processing deformation-treated samples.

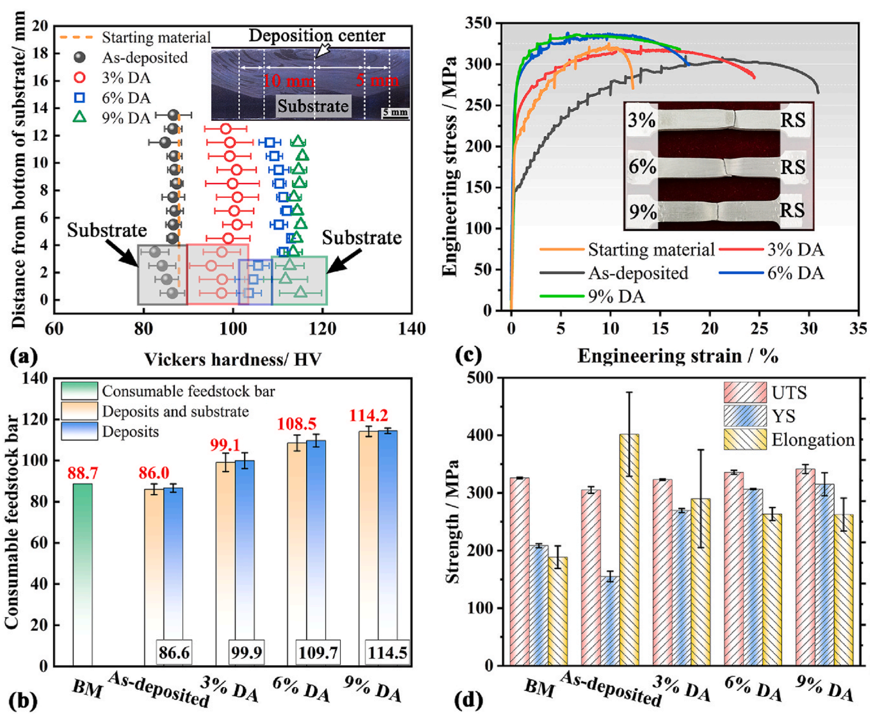

Fig. 4. Cross-sectional hardness profiles, tensile stress-strain curves, and mean value calculation of the consumable feed rod, as-deposited and post-processing deformation-treated samples. The inset in the top right corner of Fig. 4(a) illustrates the macro view cross-section of the sample after 6% plastic deformation, with hardness test lines superimposed.

Table 2
The average values of the tensile properties of the starting material, as-deposited and post-processing deformation-treated samples.

Samples	Strength/MPa		A/%	E/GPa	References
	R_m	$R_{p0.2}/R_{eL}$			
Starting material	326.0 ± 1.4	208.5 ± 3.5	13.2 ± 1.4	68	This study
As-deposited	305.3 ± 5.7	155.0 ± 9.1	28.1 ± 5.1	64	
3% DA	323.0 ± 1.4	269.5 ± 3.5	20.3 ± 5.9	65	
6% DA	335.5 ± 3.5	306.5 ± 0.7	18.4 ± 0.8	68	
9% DA	341.5 ± 7.8	315.0 ± 19.8	18.4 ± 2.0	69	
5083-H112	275	125	14	–	[16]
6061-T6	290	240	10	–	

deformation at this level. Upon evaluating both the uniformity of deformation (as indicated by the standard deviation of test values) and the improvement in tensile properties, a deformation amount of 6% was determined to be the optimal deformation process parameter.

In the present investigation, samples subjected to post-processing deformation retained the refined grain structure characteristic of the as-deposited state, even after minor compressive deformation. However, there was a notable increase in the density of GNDs, ranging from 2 to 5.6 times that of the as-deposited sample. To quantify the contribution of dislocation strengthening (Δσdis), the density of GNDs was employed in the subsequent equation, as referenced in Ref. [16]:

For the non-heat treated Al alloy 5083, solid solution strengthening, grain refinement strengthening, dislocation strengthening, and Orowan strengthening are considered to be the primary strengthening mechanisms. In this study, the deformation-treated samples retained the refined grains of the as-deposited state. In addition, they exhibited a GNDs density that was 2–5.6 times higher than that of the as-deposited sample. The GNDs density was used here to calculate the contribution of dislocation strengthening ($\Delta\sigma_{dis}$) using the following equation [17]:

$$\Delta\sigma_{dis} = M\alpha Gb\rho^{1/2} \quad (1)$$

where M is the mean orientation factor for aluminum alloy (3.06) [17], α is the constant for fcc metals (0.2) [18], G is the shear modulus of the Al-5083 (27.0 GPa) [19], b is the Burgers vector (0.286 nm), and ρ is the GNDs density. Compared with the as-deposited specimen (19.7 MPa), the contribution of dislocation strengthening of the post-processing deformation-treated samples increased to 28.0, 40.3 and 46.7 MPa, respectively, with the deformation increase. This is mainly due to the fact that the enormous number of immobile dislocations generated during plastic deformation can significantly hinder the mobile dislocations from moving and thus increase the dislocation strengthening effect.

The contribution of grain refinement strengthening at different locations can be obtained using the Hall–Petch equation [19,20]:

$$\Delta\sigma_{H-P} = k_y d^{-1/2} \quad (2)$$

where k_y is a constant, and the value is generally taken as 0.22 MPa $m^{1/2}$ for Al–4%Mg [19,20]. The contribution of grain refinement strengthening of the as-deposited sample ranged from 79.3 to 90.9 MPa. The contribution of grain refining strengthening of the deformed specimen after processing was between 78.3 and 92.1 MPa.

The incoherent second phases in the Al-5083 alloy were considered to be broken into a fragmented or chain-like distribution during the cold deformation process. The Orowan strengthening effect is inversely proportional to the distribution distance of the incoherent second particles [21,22]. Consequently, the contribution of Orowan strengthening, arising from the interaction between these fragmented, incoherent second phases and the markedly increased dislocation density, is significantly enhanced.

It can be challenging to apply a pre-stretch process to large and complex additively manufactured parts that require post-processing strengthening. Rolling or other alternative approaches that can achieve compressive deformation are deemed more suitable for practical applications. Although this study demonstrated that applying a small amount of compressive deformation could significantly enhance the strength, further evaluation of the anisotropy and corrosion resistance of parts subjected to post-processing deformation is required in subsequent studies.

4. Conclusions

This study first explored the influence of post-processing deformation on the microstructure and mechanical properties of Al-5083 produced via AFSD.

(1). Compared with the as-deposited specimen, the post-processed samples subjected to plastic deformation escalating from 3% to 9% exhibited a pronounced enhancement in dislocation generation, accumulation, and ensuing interactions. The GNDs density increased from a factor of 2–5.6, corresponding with a rise in the subgrain fraction from 20.4–34.0% to 43.7–59.8%.

(2). Regardless of deformation amount, the yield strength of post-processed deformation treated specimens rose by at least 73.9%, attributed to increased dislocation density enhancing both dislocation and Orowan strengthening. They maintained most of their ductility owing to inheriting fine grains from the deposited state.

(3). The optimum amount of deformation was 6% as the deposited layer showed greater uniformity and stability in tensile properties. The ultimate tensile strength, yield strength and the elongation at break of the deformation-treated specimen (6% strain) reached 335.5 ± 3.5 MPa, 306.5 ± 0.7 MPa and 18.4 ± 0.8%, respectively.

Originality statement

I write on behalf of myself and all co-authors to confirm that the results reported in the manuscript are original and neither the entire work, nor any of its parts have been previously published. The authors confirm that the article has not been submitted to peer review, nor has been accepted for publishing in another journal. The authors confirm that the research in their work is original, and that all the data given in the article are real and authentic. If necessary, the article can be recalled, and errors corrected.

CRediT authorship contribution statement

Wenshen Tang: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft, Writing – review & editing, Funding acquisition. **Xinqi Yang:** Conceptualization, Resources, Methodology, Data curation, Writing – review & editing, Funding acquisition. **Ruilin Wang:** Methodology, Investigation, Writing – review & editing. **Ting Luo:** Methodology, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

[1] J.H. Martin, B.D. Yahata, J.M. Hundley, J.A. Mayer, T.A. Schaedler, T.M. Pollock, 3D printing of high-strength aluminium alloys, Nature 549 (2017) 365–369, https://doi.org/10.1038/nature23894.

[2] H.R. Kotadia, G. Gibbons, A. Das, P.D. Howes, A review of laser powder bed fusion additive manufacturing of aluminium alloys: microstructure and properties, Addit. Manuf. 46 (2021), 102155, https://doi.org/10.1016/j.addma.2021.102155.

[3] L. Zhou, H. Hyer, S. Park, H. Pan, Y. Bai, K.P. Rice, Y. Sohn, Microstructure and mechanical properties of Zr-modified aluminum alloy 5083 manufactured by laser powder bed fusion, Addit. Manuf. 28 (2019) 485–496, https://doi.org/10.1016/j.addma.2019.05.027.

[4] D. Svetlizky, B. Zheng, T. Buta, Y. Zhou, O. Golan, U. Breiman, R. Haj-Ali, J. M. Schoenung, E.J. Lavernia, N. Eliaz, Directed energy deposition of Al 5xxx alloy using laser engineered net shaping (LENS®), Mater. Des. 192 (2020), 108763, https://doi.org/10.1016/j.matdes.2020.108763.

[5] Z. Wang, X. Lin, N. Kang, J. Chen, Y. Tang, H. Tan, X. Yu, H. Yang, W. Huang, Directed energy deposition additive manufacturing of a Sc/Zr-modified Al–Mg alloy: effect of thermal history on microstructural evolution and mechanical properties, Mater. Sci. Eng. 802 (2021), 140606, https://doi.org/10.1016/j.msea.2020.140606.

[6] Z. Qin, N. Kang, M. El Mansori, Z. Wang, H. Wang, X. Lin, J. Chen, W. Huang, Anisotropic high cycle fatigue property of Sc and Zr-modified Al-Mg alloy fabricated by laser powder bed fusion, Addit. Manuf. 49 (2022), 102514, https://doi.org/10.1016/j.addma.2021.102514.

[7] M.N. Gussev, N. Sridharan, Z. Thompson, K.A. Terrani, S.S. Babu, Influence of hot isostatic pressing on the performance of aluminum alloy fabricated by ultrasonic additive manufacturing, Scripta Mater. 145 (2018) 33–36, https://doi.org/10.1016/j.scriptamat.2017.10.004.

[8] H.Z. Yu, M.E. Jones, G.W. Brady, R.J. Griffiths, D. Garcia, H.A. Rauch, C.D. Cox, N. Hardwick, Non-beam-based metal additive manufacturing enabled by additive friction stir deposition, Scripta Mater. 153 (2018) 122–130, https://doi.org/10.1016/j.scriptamat.2018.03.025.

[9] W. Tang, X. Yang, C. Tian, Y. Xu, Microstructural heterogeneity and bonding strength of planar interface formed in additive manufacturing of Al−Mg−Si alloy based on friction and extrusion, Int. J. Miner. Metall. Mater. 29 (2022) 1755–1769, https://doi.org/10.1007/s12613-022-2506-4.

[10] B.J. Phillips, C. Jacob Williamson, R.P. Kinser, J. Brian Jordon, K.J. Doherty, P. G. Allison, Microstructural and mechanical characterization of additive friction stir-deposition of aluminum alloy 5083 effect of lubrication on material anisotropy, Materials 14 (2021), https://doi.org/10.3390/ma14216732.

[11] W. Tang, X. Yang, C. Tian, C. Gu, Effect of rotation speed on microstructure and mechanical anisotropy of Al-5083 alloy builds fabricated by friction extrusion additive manufacturing, Mater. Sci. Eng. 860 (2022), 144237, https://doi.org/10.1016/j.msea.2022.144237.

[12] J. Zhang, Z. Li, F. Xu, C. Huang, Regulating effect of pre-stretching degree on the creep aging process of Al-Cu-Li alloy, Mater. Sci. Eng. 763 (2019), 138157, https://doi.org/10.1016/j.msea.2019.138157.

[13] V.V. Stolyarov, L. Zeipper, B. Mingler, M. Zehetbauer, Influence of post-deformation on CP-Ti processed by equal channel angular pressing, Mater. Sci. Eng. 476 (2008) 98–105, https://doi.org/10.1016/j.msea.2007.04.069.

[14] W. Pantleon, Resolving the geometrically necessary dislocation content by conventional electron backscattering diffraction, Scripta Mater. 58 (2008) 994–997, https://doi.org/10.1016/j.scriptamat.2008.01.050.

[15] A. Kar, S. Kumar, S.V. Kailas, Developing multi-layered 3D printed homogenized structure using solid state deposition method, Mater. Char. 199 (2023), 112770, https://doi.org/10.1016/j.matchar.2023.112770.

[16] K. Anderson, J.G. Kaufman, J. Weritz, ASM Handbook: Volume 2B Properties and Selection of Aluminum Alloys, ASM International Materials Park, OH, USA, 2019.

[17] B.Q. Han, F.A. Mohamed, E.J. Lavernia, Tensile behavior of bulk nanostructured and ultrafine grained aluminum alloys, J. Mater. Sci. 38 (2003) 3319–3324, https://doi.org/10.1023/A:1025150407956.

[18] K. Ma, H. Wen, T. Hu, T.D. Topping, D. Isheim, D.N. Seidman, E.J. Lavernia, J. M. Schoenung, Mechanical behavior and strengthening mechanisms in ultrafine grain precipitation-strengthened aluminum alloy, Acta Mater. 62 (2014) 141–155, https://doi.org/10.1016/j.actamat.2013.09.042.

[19] E.L. Huskins, B. Cao, K.T. Ramesh, Strengthening mechanisms in an Al-Mg alloy, Mater. Sci. Eng. 527 (2010) 1292–1298, https://doi.org/10.1016/j.msea.2009.11.056.

[20] T. Mukai, K. Higashi, S. Tanimura, Influence of the magnesium concentration on the relationship between fracture mechanism and strain rate in high purity AlMg alloys, Mater. Sci. Eng. 176 (1994) 181–189, https://doi.org/10.1016/0921-5093(94)90974-1.

[21] B.Q. Han, D.C. Dunand, Creep of magnesium strengthened with high volume fractions of yttria dispersoids, Mater. Sci. Eng. 300 (2001) 235–244, https://doi.org/10.1016/S0921-5093(00)01781-0.

[22] D. Hull, D.J. Bacon, Introduction to Dislocations, Elsevier, 2011.